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SLEEP BREATHING PHYSIOLOGY AND DISORDERS • ORIGINAL ARTICLE

Inpatient CPAP adherence may predict post-discharge adherence in hospitalized patients screened high risk for OSA

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Abstract

Rationale Despite increased recognition of sleep disordered breathing in hospitalized patients, studies are lacking on the impact of inpatient adherence with positive airway pressure (PAP) therapy on post-discharge adherence.

Objectives To assess the predictive value of inpatient adherence to PAP therapy on post-discharge compliance and adherence.

Methods We reviewed data on individuals as part of a registry of a hospital-based sleep medicine program between August 2019 and December 2020. Consecutive patients identified as high risk for sleep disordered breathing based on our 2-tier screening process and initiated on Auto-PAP (APAP) therapy were included. Their adherence and post-discharge course were recorded. Primary objectives were polysomnography (PSG) compliance, sleep medicine clinic follow-up compliance, and 30-day adherence to PAP therapy if indicated by PSG.

Results In total, 900 individuals were screened during the study period. Of these, 281 were offered inpatient PAP therapy. Patients on bilevel PAP therapy (88 patients) were excluded due to lack of objective compliance recording. Final analysis was performed on 193 patients. Of the 193 patients placed on inpatient APAP, 140 (73%) were adherent to the therapy with average usage of 367 min per day versus 140 min per day in the non-adherent ($p < 0.001$). There was no significant difference in oxygen desaturation index between the adherent and non-adherent groups (32.4 ± 21.9 events per hour and 34.5 ± 21.9 events per hour consistent; $p = 0.5$). No demographic and anthropometric characteristics or comorbid conditions were noted. Those who were adherent to PAP therapy in-hospital 47/140 (34%) underwent ambulatory PSG post-discharge compared to 7/53 (13%) of those non-adherent in-hospital ($p = 0.002$). The adherent group also had significantly higher likelihood for post-discharge clinic follow-up ($p = 0.01$) and adherence to outpatient PAP therapy ($p = 0.01$).

Conclusions Hospitalized patients identified as high risk for sleep disordered breathing have high adherence to PAP therapy during hospitalization and inpatient adherence predicts outpatient follow-up (both PSG testing and sleep clinic) and home PAP adherence.

Keywords Hospitalized patients · Sleep disordered breathing · PAP therapy · Adherence to therapy

Introduction

Sleep disordered breathing (SDB) is a common yet under-recognized condition with significant health implications [1–4]. A recent study has suggested that a billion people worldwide have undiagnosed SDB [5]. Addressing this discrepancy will likely take a multifaceted approach including assessment of SDB in both the ambulatory setting and during hospitalization. Several studies have demonstrated the utility of screening for sleep disordered breathing during hospitalization [6–8] . Hospital sleep medicine programs offer a unique opportunity to intervene on this historically outpatient disorder as these patients represent a "captive audience." In fact, studies have demonstrated that proactive

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screening for SDB in hospitalized individuals reduces readmission rates and mortality [9–11]. An interesting yet unstudied facet of hospital sleep medicine is the impact of treatment with positive airway pressure (PAP) therapy during the hospital stay. The impact of inpatient therapy may be of clinical significance as a predictive tool in an environment of limited healthcare resources.

Our local institution has adapted a well-described hospital sleep medicine program previously described by Sharma and colleagues [12, 13]. This tiered screening program identifies patients at high risk for SDB upon entry into our system. Patients are educated concerning the disease process and its impact on comorbid health conditions. Recommendations are provided during the patient's hospital stay regarding the appropriate screening test and there is coordination with care management in the discharge process. An important aspect of this hospital sleep medicine program is a trial of PAP therapy in those identified at particularly high risk for SDB. The details of this program are published elsewhere [13] . Evidence suggests that patients form and establish beliefs related to PAP therapy during the first 72 h of treatment [11]. Thus, this initial introduction of PAP therapy is likely of critical long-term importance and may have practical implications in terms of hospital-based screening for SDB and post-discharge care for these patients.

In this study, we aimed to assess the predictive value of inpatient adherence to Auto-CPAP (APAP) therapy on [1] polysomnography (PSG) follow-up post-discharge; [2] sleep medicine clinic follow-up; and [3] outpatient adherence to CPAP at 30 days after APAP initiation if this treatment was deemed appropriate after outpatient confirmatory testing. We hypothesized that individuals adherent to therapy during their hospital stay would be more likely to follow-up with outpatient sleep testing and sleep medicine clinic appointments, and be more adherent to PAP therapy at 30 days.

Methodology

Our single center study reviewed data from individuals as part of a clinical hospital-based sleep medicine program. Participants were those patients enrolled in the screening program from August 2019 to December 2020. The program's tiered screening process identifies all obese patients with a BMI $\geq$ 30 kg/m^2 admitted to an inpatient medical service via the electronic medical record (EMR). These identified individuals are then initially screened by the sleep medicine team using the STOP/ STOP-BANG questionnaire and the Epworth Sleepiness Score [14, 15] . Positive initial screens were followed up by a secondary screening study using high-resolution pulse oximetry (HRPO) or a portable sleep monitoring device (PM) to determine respiratory event indices by either oxygen desaturation index (ODI) or apnea–hypopnea index (AHI) [16, 17] . The data from these secondary studies were reviewed by a board-certified sleep medicine physician. Those with an ODI or AHI $\geq$ 15/h were considered to have an elevated risk for SDB; they were offered initiation of inpatient APAP therapy under the direction of the sleep medicine consult service which included a board-certified sleep medicine physician. This therapy was subsequently maintained throughout their hospitalization. In-depth description of this screening protocol is available elsewhere [12, 13] . Upon discharge, patients were advised to follow up at the sleep disorders center and undergo a confirmatory PSG study. All patients underwent PSG instead of home sleep testing due to underlying cardio-pulmonary comorbidities including CHF and COPD.

Our study team analyzed inpatient adherence to APAP therapy and the likelihood of follow-up with PSG, clinic follow-up, and outpatient adherence to PAP therapy (if ultimately determined to be necessary through outside confirmatory testing). While there is no clear definition for optimal inpatient adherence to PAP therapy, our team defined "adherent" as consistent use of PAP therapy for $\geq$ 4 h per day as objectively recorded by the APAP device based on the current rules defining PAP adherence in the outpatient setting [18] . Outpatient adherence was defined as $\geq$ 4 h of nightly use for 70% or more of the nights. Only patients placed on APAP therapy were considered in this trial due to the capability of the device to record the amount of usage. This study was approved by the WVU IRB (# 2,009,120,369).

Statistical analysis

Group comparisons were used to analyze differences between those adherent and those non-adherent to therapy. Student's *t* test was used for continuous variables; the Fisher exact test was used for dichotomous variables. Probability levels of $p \leq 0.05$ were considered significant. Multivariate logistic regression analysis was performed using data from all 193 individuals placed on APAP therapy to further assess factors informing outpatient outcomes.

Results

As shown in Fig. 1, a total of 900 patients (53% men) were screened by HRPO or PM as part of the clinical screening program. Of these, a total of 281 individuals were offered inpatient PAP therapy. The main reasons for exclusion were a negative screen for moderate-to-severe SDB (thus, PAP was not indicated), or an early discharge prior to initiation of therapy. Eighty-eight of the 281 individuals initiated on PAP therapy were determined by the attending physician to have an indication for bilevel therapy and were excluded due to lack of objective

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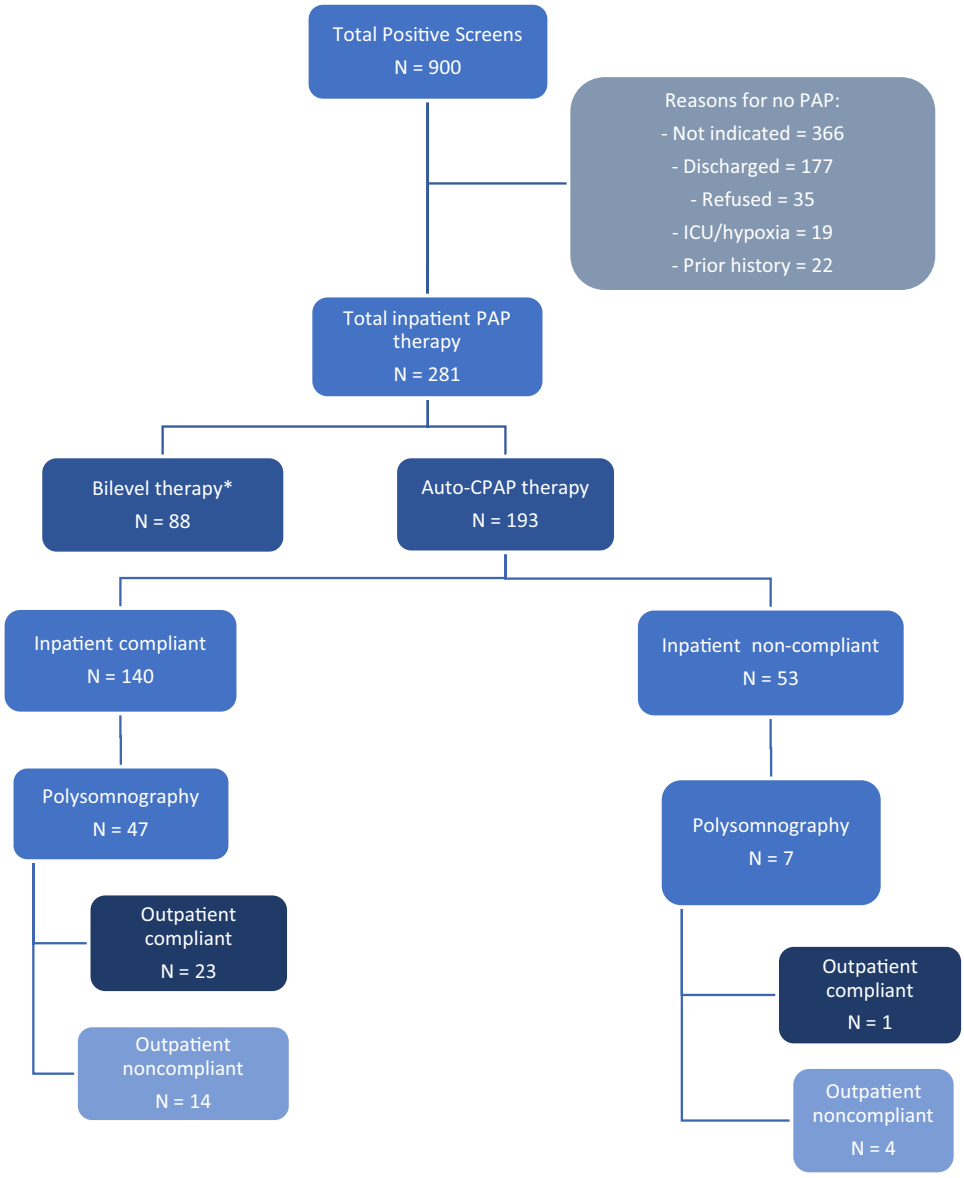

Fig. 1 Flow chart showing recruitment of patients. *Excluded as device unable to record compliance. Auto-CPAP, auto continuous positive airway pressure

*excluded as device unable to record compliance

Auto-CPAP – Auto continuous positive airway pressure

compliance recording on our devices. The indications for bilevel therapy included intolerance to APAP, concomitant hypercapnia, or non-availability of an APAP device. Therefore, analyses were performed on the 193 patients who were placed on APAP and who had objective adherence data available. The mean age of the group ($n = 193$) was 57.1 ± 12.2 years (SD), mean BMI was 41.5 ± 14.1 kg/m^2 (SD), and 69% were men. There was no significant difference in ODI between the adherent and non-adherent groups (32.4 ± 21.9 events per hour and 34.5 ± 21.9 events per hour consistent; $p = 0.5$). There was no significant difference in AHI between the adherent and non-adherent groups respectively (31.9 ± 32.1 events per hour and 30.8 ± 18.1 events per hour; $p = 0.8$). As part of another ongoing project, some patients had both an HRPO and PM completed on the same evening. Both ODI and AHI were recorded as part of our analysis.

Of the 193 patients placed on inpatient APAP, 140 (72.5%) were adherent to the therapy with average usage of 367 min per day versus 140 min per day in the non-adherent ($p < 0.001$). Differences in demographics and baseline characteristics between the adherent and non-adherent group in the hospital are shown in Table 1. There were no differences between the two groups with respect to any of the demographic or anthropometric characteristics, or comorbid medical conditions.

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Table 1 Demographic and baseline characteristics between the adherent versus non-adherent group

	Adherent	Non-adherent	*p* value
N	140	53	
Age	56.8 ± 13.2	57.8 ± 11.9	0.56
Gender	66%	73%	0.47
BMI (kg/m^2)	40.9 ± 13.5	43.1 ± 15.8	0.51
Neck circumference (inches)	18.3 ± 2.6	18.6 ± 2.5	0.53
Health literacy	20.9 ± 5.1	20.7 ± 5.0	0.62
Length of stay (days)	8.6 ± 9.2	8.6 ± 6.1	0.99
Left ventricular ejection fraction	47.9 ± 18.4	48 ± 21.2	0.97
Hypertension	73%	82%	0.24
Congestive heart failure	55%	58%	0.82
Atrial fibrillation	30%	33%	0.81
Chronic obstructive pulmonary disease/asthma	33%	20%	0.12
Pulmonary hypertension	22%	22%	1
Diabetes mellitus type 2	48%	56%	0.36
Cardiomyopathy	25%	24%	0.95
Coronary artery disease	38%	33%	0.57
Cerebral vascular accident	12%	9%	0.8

Table 2 shows SDB characteristics and results of subsequent outpatient follow-up in the inpatient adherent and non-adherent groups. Adherent patients had a higher mean Epworth Sleepiness Score compared to non-adherent patients (10.8 ± 5.1 versus 8.7 ± 5.2; $p = 0.02$). Average pressure use was higher in the adherent group (10.7 ± 3.6 cm H_20 versus 9.1 ± 3.7 cm H_20; $p = 0.008$). Severity of SDB on screening, time with oxygen desaturation below 88%, BMI, and prevalence of medical comorbidities also were not different between groups.

As shown in Table 2, of those who were adherent to PAP therapy in-hospital, 34% (47/140) underwent confirmatory PSG post-discharge compared to 13% (7/53) of those non-adherent in-hospital ($p = 0.002$). Similarly, patients adherent to PAP therapy during hospitalization had a significantly higher likelihood of post-discharge sleep clinic follow-up ($p = 0.01$) and were more likely to be adherent to outpatient PAP therapy ($p = 0.01$).

Table 3 shows additional characteristics of those adherent to PAP as an inpatient in comparison to those who were non-adherent stratified by subsequently having a follow-up PSG, attending a follow-up clinic appointment, or being compliant with APAP after 30 days of initiation of treatment. Patients having a follow-up PSG also had a higher pre-treatment Epworth Sleepiness Scale score and shorter hospital length of stay (LOS) compared to the patients who did not follow up for PSG. Patients attending a follow-up clinic appointment were older and had a shorter LOS and lower residual AHI on APAP compared to patients who did not follow up in the clinic. Compliance after 30 days additionally was marginally associated with a shorter LOS and lower residual AHI. Multivariate logistic regression analyses (Table 4) confirmed that inpatient adherence to APAP was the strongest predictor of subsequent PSG and clinic follow-up as well as remaining compliant with APAP. Shorter length of stay remained a significant factor for completing a subsequent PSG and attending a clinic follow-up but was not associated with outpatient APAP compliance. The effects of residual AHI and Epworth Sleepiness Scale were no longer significant in these models, but older age still contributed to the likelihood of a clinic follow-up. Table 4 shows odds

Table 2 Sleep-related outcomes between the adherent and non-adherent group

	Inpatient adherent	Inpatient non-adherent	*p* value
N	140	53	
Epworth	**10.8**	**8.5**	**0.02**
STOP-BANG score	5.8 ± 1.6	5.6 ± 1.4	0.5
HRPO tested	124 (89%)	48 (91%)	
HRPO ODI	32.4 ± 21.9	34.5 ± 16.7	0.5
HRPO time spent < 88%	75.9 ± 100.4	59.3 ± 66.5	0.2
Apnea link tested	105 (75%)	39 (74%)	
Apnea link AHI	31.9 ± 32.1	30.8 ± 18.1	0.79
Average use (min)	**367.3**	**139.8**	**< 0.001**
Average leak	27.5 ± 28.3	38.1 ± 41.3	0.14
Residual AHI	11.9 ± 13.5	13.4 ± 14.7	0.53
Average pressure	**10.7 ± 3.6**	**9.1 ± 3.7**	**0.008**
Outpatient PSG completed (N)	**47 (34%)**	**7 (13%)**	**0.002**
Outpatient PSH AHI	38.4 ± 28.1	30 ± 17.1	0.26
Outpatient follow-up	**37 (26%)**	**5 (9%)**	**0.01**
Outpatient compliant	**23 (16%)**	**1 (2%)**	**0.01**

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Table 3 Characteristics of inpatient PAP adherent patients returning for polysomnography, returning for clinic follow-up, and compliant with APAP after 30 days versus patients non-compliant or not returning for follow-up

Characteristic	PSG completed		Clinic follow-up		Outpatient APAP compliant	
	Yes	*No*	*Yes*	*No*	*Yes*	*No*
N	47	93	37	103	23	117
Age	58 ± 13	57 ± 12	60.5 ± 13.9	56.1 ± 11.6[a]	57 ± 12	57 ± 12
Gender, % men	63	74	67	70	71	69
BMI, kg/m^2	40.3 ± 13.1	41.9 ± 14.5	39.0 ± 9.3	42.1 ± 15.2	37.8 ± 9.9	42.0 ± 14.6
Epworth Sleepiness Scale score	11.4 ± 5.6	9.7 ± 4.9[a]	10.3 ± 4.7	10.2 ± 5.3	9.7 ± 4.1	10.3 ± .3
Neck circumference, cm	18.4 ± 2.5	18.3 ± 2.6	18.3 ± 2.1	18.4 ± 2.7	18.1 ± 2.3	18.4 ± 2.6
Health literacy	21.0 ± 4.6	20.8 ± 5.2	20.3 ± 5.1	221.0 ± 5.0	20.5 ± 5.8	20.9 ± 5.0
Length of stay, days	6.2 ± 3.7	9.6 ± 9.6[a]	5.7 ± 3.4	9.4 ± 9.2[a]	5.9 ± 3.4	9.0 ± 8.9[b]
Left ventricular ejection fraction	49.5 ± 18.8	47.3 ± 19.3	52.0 ± 16.4	46.8 ± 19.7	51.9 ± 15.3	47.4 ± 19.6
Residual AHI, no./h	10.7 ± 11.3	13.0 ± 14.8	8.4 ± 9.5	13.5 ± 14.7[a]	7.3 ± 8.8	13.1 ± 14.3[b]
HRPO ODI, no./h	34.1 ± 19.6	32.6 ± 20.9	35.1 ± 20.9	32.5 ± 20.5	33.7 ± 18.9	32.9 ± 20.8
Apnea link AHI, no./h	33.9 ± 21.1	30.8 ± 31.1	30.7 ± 19.5	31.8 ± 30.9	31.1 ± 17.2	31.7 ± 30.0
Hypertension, %	72	71	74	76	75	75
Heart failure, %	50	58	48	58	42	58
Atrial fibrillation, %	28	31	26	32	21	32
COPD/asthma, %	32	28	29	29	25	30
Pulmonary hypertension, %	15	24	19	23	17	23
Diabetes, %	48	51	50	50	38	52
Cardiomyopathy, %	17	28	14	28	17	26
Coronary artery disease, %	20	25	38	36	29	38
Stroke, %	9	12	10	11	13	11

[a] $p < 0.05$
[b] $p = 0.09$

Table 4 Multivariate logistic regression models predicting outpatient PSG, clinic visit, and APAP compliance

	β	*Odds ratio*	*95% CI*	
			Lower	*Upper*
Polysomnography				
Epworth Sleepiness Scale	0.060	1.062	0.991	1.137
Length of stay	−0.076	0.927	0.86	0.997
Inpatient compliant	0.989	2.689	1.027	7.039
Clinic follow-up				
Age	0.036	1.037	1.004	1.071
Length of stay	−0.104	0.902	0.817	0.995
Residual AHI	−0.034	0.967	0.933	1.002
Inpatient compliant	1.230	3.420	1.213	9.643
APAP compliance				
Length of stay	−0.063	0.939	0.853	1.033
Residual AHI	−0.043	0.957	0.910	1.007
Inpatient compliant	2.284	9.814	1.275	75.546

of outpatient PSG, outpatient sleep clinic follow-up, and ambulatory adherence to PAP therapy in patients adherent in-hospital.

There were a total of 29 patients in this cohort with predominant central sleep apnea/Cheyenne Stokes Breathing (CSA/CSR). These patients were placed on a fixed pressure between 6 and 10 cm of H_2O pressure as tolerated. Of the 29 patients, 19 were compliant on PAP therapy. The mean AHI of compliant patients was 55.2/h and residual AHI was 23.4/h. In the non-compliant group of 9 patients, the mean baseline AHI was 35.4/h and the mean residual AHI was 20.2/h.

Discussion

This is the first study to analyze systematically the predictive outcome of APAP initiation and adherence in hospitalized patients screened high risk for sleep disordered breathing. Our study shows that patients who screened positive for SDB in the hospitalized setting have high adherence to PAP therapy (74%) during their hospital stay. Importantly, patients who are adherent to PAP therapy during hospitalization also have a higher likelihood of follow-up for confirmatory PSG testing post-discharge and sleep clinic appointments. Patients screened as high risk for SDB during their

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hospital stay and who were adherent to APAP therapy during hospitalization also have higher ambulatory PAP adherence at 30 days. These findings have significant implications for care.

Data have demonstrated that patients screened for sleep apnea during hospitalization and who are ultimately successfully treated with PAP therapy have significant improvement in hospital readmission rates and potentially mortality [9–11] . We now show that APAP adherence during hospitalization predicts outpatient adherence to PAP therapy. This is consistent with our prior data which showed hospitalized heart failure patients diagnosed with SDB have adherence similar to those diagnosed in ambulatory settings [19] . This important finding could have economic implications not only for decreasing cost related to readmission in those adherent to therapy but also in allowing more targeted deployment of scarce healthcare resources to those individuals at higher risk for readmission.

Our multivariate analyses assessed the relative contributions of different variables to the outcomes of return to clinic and follow-up for PSG. While variables such as length of stay had a marginally (but statistically significant) impact on follow-up, inpatient adherence to APAP therapy by far was the strongest predictor of outpatient clinical follow-up and completion of a PSG. These results, along with the finding of higher ambulatory PAP adherence in those adherent to APAP during hospitalization, underline the importance of this first introduction of PAP therapy to patients during their hospital course.

It was interesting to note that despite SDB not being the primary reason for admission, most patients accepted and adhered to the therapy (72.5%). Both adherent and non-adherent patients had similar demographic profiles and comorbidities. The only difference was an elevated Epworth Sleepiness Scale (ESS) in patients with better adherence. This is consistent with prior data showing ESS > 10 to be an independent factor in predicting long-term use of PAP therapy [20, 21] . Higher PAP pressures requirements were also noted in the patients who were adherent to the therapy. Despite the higher pressure requirement in the adherent group, mask leak was similar to the non-adherent group. While significant improvements in mean ODI/AHI were noted in both groups, there was still a small elevation in AHI. This small difference in residual AHI is unlikely to be clinically relevant. Our mean BMI was quite high at 41.5 ± 14.1 kg/m^2 which is consistent with our geographic region. All patients were screened for obesity hypoventilation and a few required intervention with bilevel PAP or average volume-assured pressure support and were excluded from the study.

Finally, our multivariate regression models support inpatient APAP adherence as the most important variable in predicting outpatient PSG and clinic follow-up. The only other significant variable found in both models was shorter hospital LOS which also was associated with a greater likelihood of follow-up. The explanation for this finding is unclear. However, one possibility is that individuals with a longer LOS may have a more complex post-discharge course including discharge to a care facility and more post-hospital testing and follow-up as well as more deconditioning related to the prolonged hospital course, thus making follow-up more challenging. Quite simply these individuals may have burnout from excessive healthcare exposure. This finding warrants further investigation.

Hospital sleep medicine is an appealing concept considering the significant under-recognition of sleep apnea and health implications if left untreated [1, 22] . In addition to high prevalence of SDB in hospitalized patients, they also are a “captive audience” [23] . This provides more opportunities for education, mask selection, and acclimatization trials. However, data on outcomes including readmissions and mortality suggest that this venue of intervention requires sustained effort. This first study analyzing APAP therapy adherence during hospitalization and outpatient outcomes suggests adherence to treatment improves outpatient follow-up.

While our findings are compelling, a few study limitations should be noted. Not all eligible patients received APAP and hence had to be excluded (for instance, patients on bilevel therapy). Adherence on APAP therapy was calculated by the number of hours used per 24 h. This study was completed at a single center that cares primarily for rural-dwelling individuals who lack access to care including sleep medicine providers and specialty sleep centers. This fact may bias the results as these individuals may be less likely to have had a SDB assessment in the ambulatory setting compared to more affluent urban populations. In addition, the study population was relatively small; while our findings were statistically significant and clinically meaningful, confidence intervals were wide and thus, there may be some imprecision in the estimates of the odds ratios. Patients non-adherent to PAP therapy in the hospital may in general be non-adherent. Finally, patients were followed up for only 30 days; longer follow-up would have been both desirable and informative. Nevertheless, this study sets the groundwork for a larger multicenter study evaluating more diverse populations with longer follow-up.

Conclusion

Our study suggests that hospitalized patients with positive screens for OSA have a high adherence to PAP therapy during the acute illness, and that inpatient adherence predicts outpatient follow-up (both PSG study and sleep clinic) and home PAP adherence. This may also allow identification of

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patients who would be "high yield" to follow-up in ambulatory settings in this cost-conscious environment in addition to protecting them from bad outcomes during their hospital stay.

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Data availability The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee (name of institute/committee) and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Consent to participate For this type of study, formal consent is not required.

Conflict of interest Dr. Stansbury serves on the clinical advisory board for ResMed Inc. and receives a consulting fee. Dr. Quan reports research funding from the National Institutes of Health (NIH), serves as a consultant to Jazz Pharmaceuticals, WhisperSom, Bryte Bed, and DR Capital and is a committee chair and a taskforce member for the American Academy of Sleep Medicine. Dr. Sharma is a task force member of the American Academy of Sleep Medicine. No other authors report any conflict of interest.

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